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44:43 spoken · 51 beats · 121 wpm · generated by tools/print-script.py

frame 1 · title · A0.1 · 0:27 · 0:27 in

Thank you, and good afternoon. Thank you all for coming, and thank you to the committee for reading this and being here in person.

I'm very happy today to present the work that I have been doing for my PhD over the past 3 years, on methods for trustworthy non-Gaussian inference for weak-lensing cosmology.

frame 2 · the picture · A0.2 · 0:25 · 0:52 in

I would like to start with the the history of the Universe as we currently model it.

It starts in a hot, almost uniform state, with very small fluctuations in density. Then it expands. And those fluctuations grow under gravity into the structure we observe today — galaxies, clusters, the cosmic web.

frame 3 · what ΛCDM is, and where it stops · A0.3 · 1:19 · 2:11 in

Our prevailing model describing all of that is ΛCDM — Lambda, cold dark matter. It is a very simple model, with just six free parameters.

It rests on this short list of assumptions, basically that gravity is given by general relativity, that the Universe is homogeneous and isotropic on large scales, that the initial conditions come from inflation, and are near-Gaussian and adiabatic, and that the matter is a mixture of baryons, cold dark matter and a cosmological constant.

▲ And it has been extremely successful. Those six numbers fit our different observations to a remarkable precision.

[CLICK] ▲ But it is still not a fully satisfactory answer, for a few reasons. First of all, it is a **phenomenological** description rather than a first-principles explanation. About ninety-five per cent of what it describes is two **dark components** whose nature we do not know. And independent probes of the same parameters have started to show **tensions** — statistically significant disagreements in the cosmological parameters they report.

frame 4 · the probes, and the one we follow · A0.4 · 0:16 · 2:27 in

So how do you study the Universe, and test a model like that? Well, there are **several probes** that do that...

[CLICK] ▲ But the one we follow in this thesis is **gravitational lensing**.

frame 5 · strong and weak lensing · A0.5 · 0:46 · 3:13 in

Here is the basic idea of gravitational lensing.

Light from a distant source passes through all the mass distribution on the way to us, and the gravity of this mass **bends its path**, so the image arrives distorted.

Occasionally that is dramatic — arcs, multiple images, Einstein rings, visible by eye. That is the **strong** regime, and it is *very rare*.

▲ What happens *everywhere* is the **weak** regime: every galaxy behind any structure has its shape slightly changed. This gives us an indication of how the matter (including the dark matter) is distributed.

frame 6 · Euclid, and what the signal is · A0.6 · 0:58 · 4:11 in

And we are about to be able to do this to a new level of precision. The next generation of cosmological surveys is here, and the one I work on is the **Euclid** space telescope.

Euclid launched in 2023, it is taking data now, and it will measure the shapes of **billions of galaxies** over **a third of the sky**, giving us roughly an *order of magnitude* more statistical power than anything we have.

[CLICK] ▲ That signal encodes the growth of structure and the geometry of the Universe since the Big Bang.

But in order to extract this information and do cosmology with it we need really sophisticated **algorithms and statistics**.

That is what this thesis is about.

frame 7 · the chain, and the two halves · A0.8 · 1:22 · 5:33 in

Modern cosmological survey analyses are **incredibly complex**. But here is the chain reduced to just the last steps, which are the scientific part of the analysis.

Galaxy shapes go in. From the shapes we make a **map of the mass distribution**. From the map we extract **a few numbers, the summary statistics**. We compare those with theoretical predictions, or with simulations, and we get the probability distributions of the parameters.

[CLICK] ▲ The first half of the talk, the first two papers, is about this step: **making the map**.

[CLICK] ▲ The second half, the last two papers, is about everything after it: extracting information from the the map, and using it to infer cosmology.

Every one of these steps can bias the result or distort the error bars, if it fails to capture the relevant physical or observational effects. That's why for the **results to be trustworthy**, we need methods that properly quantify the unceirtainty, are calibrated, and tested for being unbiased. Otherwirse, we risk producing incorrect scientific conclusions.

Act 1 — Part 1, does the map matter? · frames 8–20

frame 8 · the question · A1.1 · 0:06 · 5:40 in

So.. Let's start with **our first paper**.

First, what is exactly being reconstructed.

frame 9 · shear and convergence · A1.2 · 0:50 · 6:29 in

The effect of weak lensing on the shapes of the galaxies can be summarised in two quantities: the convergence, which is the **an isotropic magnification** of the image, and the shear, which is **the anisotropic stretching** of the image.

Convergence is **not directly observable**

But the shear is measurable through statistical analysis of the shapes of many galaxies. ▲ The idea is that the lensing is **coherent** and to a good approximation, the intrinsic shapes are **randomly distributed**. Therefore, if we average the ellipticities over many galaxies in a patch, the random part cancels, and what remains is the pure shear.

`frame 10` · `what the convergence is · A1.3 · 1:01 · 7:30 in`

However, we would really like to know the convergence, for a few reasons.

First of all, it is a **scalar field**, so it is much easier to work with and extract information from than the shear which is a spin-2 field. In fact most non-gaussian statistics which we will talk about later are only defined on convergence.

Second, it has a direct physical interpretation: the convergence is the **projected matter along the line of sight**, weighted by how efficiently each piece of it lenses. So it essentially tells us the distribution of matter in the Universe. That's why convergence maps are also called **mass maps**.

Here you can see an example of a real mass map I created for the **UNIONS** galaxy survey.

`frame 11` · `shear and convergence from the same potential · A1.4 · 0:26 · 7:57 in`

Lucky for us, the shear and convergence are not independent. They are both second order derivatives of this quantity called the **lensing potential**, which is itself a projection of the gravitational potential along the line of sight.

In Fourier space we can go from one to the other with a **simple linear relation**.

`frame 12` · `the relation is exact, the measurement is not · A1.5 · 0:43 · 8:39 in`

While the relation is exact, the data is **imperfect**. The shear is measured from galaxy shapes, which are irregularly sampled, and have noise (much larger than the shear itself) and *masks* -- missing regions where we have no data, or the data is not usable.

This is why mass mapping is an **ill-posed inverse problem**: there are multiple possible solutions visibly different from each other and consistent with the data.

`[CLICK]` ▲ To get a unique solution, we have to make an assumption about what the kappa looks like.

`frame 13` · `Kaiser–Squires · A1.6 · 1:00 · 9:39 in`

The simplest method, and the standard one, is Kaiser–Squires: which simply applies the linear inversion directly to the measured shear. Almost every survey has used it for thirty years, mainly for its simplicity.

`[CLICK]` But on real data (as you can see in this comparison here) it **amplifies the shape noise** at small scales, and also causes **leakage of the masked areas** across the whole map, because the operator is non-local.

In fact, the actual "pure" Kaiser-Squires reconstruction is so noisy that it is almost impossible to discern features in. So what we do in practise, and what we have done in this image is to apply **gaussian filtering**, which reduces the noise, at the expense of *removing the smaller scales*.

`frame 14` · `a data term plus a regulariser · A1.7 · 0:38 · 10:17 in`

But moving towards more advanced methods, every method can be written as an optimisation problem with two terms: a **data-fidelity term**, how well the map reproduces the measured shear; and a **regulariser, which encodes the prior** — what we assume a plausible convergence map looks like.

The data term is the **same for all of them**. What changes from one method to the next is *the prior they assume for* κ, and the way the optimisation is solved.

`frame 15` · `MCALens · A1.9 · 0:51 · 11:08 in`

Among **model-driven** methods — those with a hand-crafted prior, rather than one learned from data — the state of the art is **MCALens**. It models the convergence as the sum of two components: **a Gaussian component**, estimated with a Wiener filter; and a **non-Gaussian component**, representing the peaks in the field The two are estimated by **alternating minimisation**: solve for one holding the other fixed, then the other, and iterate until the algorithm converges.

`[CLICK]` Each sub-problem is solved with what is called a "**proximal step**": a gradient step towards the measured shear, then an operator that pulls the result back to what the prior allows.

`frame 16` · `the question · A1.10 · 0:49 · 11:57 in`

So different priors give different maps, all consistent with the measured shear. Until this paper, mass-mapping methods were compared just on the **quality of the map**: how close the reconstruction comes to the truth in terms of mean-square error, in simulations. But in practice,the map is *not the final product*. What we care about is the **cosmological parameters**, and the reconstruction error does not tell us how the choice of method affects them.

`[CLICK]` For cosmological surveys like Euclid this is a practical question: is an advanced reconstruction method **worth the effort**, or is any reasonable method good enough?

`frame 17` · `the experiment · A1.11 · 0:37 · 12:34 in`

To answer it, we built a pipeline in which everything is **held fixed** except the mass-mapping method: we work with the **cosmo-SLICS** simulations, on a Euclid-like setup, and we run the whole chain from shear to posterior for each method. These details are not that important right now, I will come back to all these steps in Part 3 and Part 4.

`[CLICK]` What matters is that **the only thing that varies** is the reconstruction method.

`frame 18` · `the three maps · A1.12 · 0:57 · 13:31 in`

Here are the three example reconstructions of the same simulated field, next to the truth. The methods that we use as baseline are Kaiser–Squires, and a version of Kaiser–Squires with *inpainting*: a technique that fills the masked areas in a way that is matching statistically the existing field, and is supposed to reduce the mask leakage issues.

These two are the methods that are planned to be used in **Euclid**, where mass mapping has been treated as preprocessing with little effect on the cosmology, so the simplest method became the default.

MCALens on the other hand is our **state of the art method**.

So what are the posteriors that these lead to, in our work?

`frame 19` · `the answer · A1.13 · 1:41 · 15:12 in`

Here is the result. This is the **first corner plot** I am showing, so let me say what it is.

Each of these contours is the joint distribution of a *pair* of parameters; each curve on the diagonal is one parameter on its own. The two shades represent the **sixty-eight and ninety-five per cent** probability mass — *the smaller the contour, the more precise the measurement.*

The first thing we notice is that **inpainting makes no difference**: the grey and the red contours are almost identical — in fact the grey is hard to see at all behind the red.

But the more important thing is that **MCALens is visibly tighter**, on every parameter.

On the right, both stems are **improvement over Kaiser–Squires**, so higher is better for both. The **open** diamonds are the improvement on the map itself, the RMSE, and MCALens is **four per cent** better. The **filled** diamonds are the constraining power: a factor of **two point six**, **160 %**.

So we see that map quality and constraining power are *not the same objective*. And that by using an advanced mass mapping method we get a huge gain in precision.

So WL surveys should use advanced reconstruction methods to maximise their scientific outputs.

`frame 20` · `where the gain comes from · A1.14 · 0:46 · 15:58 in`

Thanks to the way that we implemented our analysis, we could really look into how the reconstruction of the different spatial scales affects the constraining power.

And what we find is that if we look at the information coming from smaller and smaller scales in the map, we see that the Kaiser–Squires saturates (stops improving) below **eight arcminutes**, while with MCALens the more small-scale information we add, *the better the constraints get*.

What this means is that MCALens is much better at recovering the **small-scale structure**, which KS loses to the noise.

Act 2 — Part 2, PnPMass · frames 21–24

`frame 21` · `the divider · A2.1 · 0:22 · 16:19 in`

So since we have established that the reconstruction method has a singnificant impact on the cosmological results, this gives us the motivation to build the best one we can for our surveys.

This is what we develop in this paper, led by **Hubert Leterme**.

`frame 22` · `what we want from a method · A2.2 · 0:45 · 17:04 in`

The obvious place to look is **deep learning** — and people have already tried it, with several approaches, and it works. But what do we want from a method, in order to use it in a survey like Euclid? **Four things**: accurate; flexible, meaning the same trained model works when the noise or the mask changes; fast; and with fast uncertainty quantification (we want error bars of the reconstruction, not just the map).

And really, **none of the existing methods have all four**. That is what we set out to build.

`frame 23` · `plug-and-play · A2.3 · 2:04 · 19:08 in`

Our method is based on the **plug-and-play framework**, from the optimisation literature. It is a way to solve an inverse problem with a learned prior, and it is very flexible.

The starting point is the iteration from Part 1: a gradient step on the data term, towards the measured shear, then a proximal step that enforces the prior. Plug-and-play replaces the proximal operator (which enforces the prior) with a **deep denoiser** trained on simulations. The iteration converges to a **fixed point**.

Why this is better than our analytical priors of e.g. MCALens? Because the denoiser is trained on simulations, it can learn a much more complex and realistic prior than what we could express analytically.

The denoiser is a Swin transformer neural network with seven million parameters, trained once on white Gaussian noise over a range of levels. ▲ With this choice of gradient step, the noise covariance and the mask enter only in the data term, at inference time, so the same network applies to any noise level and any footprint. That is what **makes it flexible**.

Here it is, visualised. We start from the two components of the shear and initialise the mass map at zero; the **forward step**, the gradient step on the data term, gives a noisy map, and then we plug in the **backward step**, the denoiser, which projects the solution to the manifold of fields that look like a **realistic mass map**.

`[CLICK]` The output is fed back, and we iterate *until the algorithm converges*.

`frame 24` · accurate, with calibrated uncertainties · A2.7 · 1:30 · 20:38 in

Not only does our method produce accurate maps, it also produces error bars for the reconstruction, which are also **perfectly calibrated**, with mathematical guarantees. We implemented this via moment neworks and a statistical technique called **conformalized quantile regression**.

And here are the results: on the x-axis we plot the reconstruction error, and on the y-axis we plot the size of the calibrated error bars, so **lower left is better**.

The **colour** is the miscoverage rate: before calibration the circles sit above the target, after calibration every diamond is on it. So the comparison is not *who covers* — everyone covers, that is the conformal guarantee doing its job — it is who covers with the **tightest bars**.

▲ On accuracy, PnPMass is within about **one per cent of DeepMass**, a deep learning method trained specifically for this mask and noise level. On uncertainty, it has the **smallest calibrated error bars** of all methods.

[CLICK] And remember, DeepMass needs retraining for every new mask or noise level, making it not applicable to an actual large survey, whereas PnPMass is *trained once and works for any field.*

Act 3 — Part 3, the summaries · frames 25–44

`frame 25` · the second half · A3.1 · 0:32 · 21:10 in

The second half of the talk is about what comes after the map: the **summary statistic** we extract from it, and the inference of the cosmological parameters from that statistic.

Notice what we have *not* touched yet. In both parts so far the summary statistic was **held fixed on purpose**, so that nothing that moved could be blamed on it. Now it is the subject.

`frame 26` · the map has to be compressed · A3.2 · 0:32 · 21:42 in

So back to our pipeline. We have a map, and we want to compare it with theory to infer the cosmological parameters. The map is of the order of a **hundred thousand correlated pixels**, so it would be extremely hard to work with directly. **[CLICK]** Instead, we compress it into a **low-dimensional summary statistic**. **[CLICK]** And the choice of statistic *determines how much information is retained.*

`frame 27` · the two-point function · A3.3 · 0:52 · 22:34 in

The standard statistic for weak lensing is the **two-point correlation function**: how correlated the shear is between pairs of galaxies separated by an angle theta, measured in real space as xi-plus and xi-minus, or in harmonic space as the power spectrum.

[CLICK] Every flagship cosmic shear result of the last decade, KiDS, DES, HSC, is a two-point analysis, and for good reasons. It can be predicted analytically, its covariance is understood, and there are two decades of work on its systematics.

[CLICK] ▲ For a Gaussian random field, the power spectrum is a **sufficient statistic**: it contains all the information. But the late-time density field is *not Gaussian.*

`frame 28` · same power spectrum · A3.4 · 0:34 · 23:07 in

Gravitational collapse and nonlinear structure formation makes the field non-Gaussian: matter concentrates into haloes and filaments around voids.

And here you can clearly see the issue with that: these two completely different fields have **the same power spectrum**.

A two-point analysis *cannot distinguish them.*

So to access the information that the power spectrum misses, we need **higher-order statistics**: peak counts, wavelet statistics, the ℓ_1-norm, Minkowski functionals, and others.

`frame 29` · the gain · A3.6 · 0:13 · 23:20 in

And this information seems to be quite valuable: from our forecasts, done on simulations, adding different higher-order statistics to the power spectrum **significantly improves the constraints**.

`frame 30` · wavelets · A3.10 · 1:28 · 24:48 in

But to talk about the higher-order statistics we investigate and develop in this thesis, we need to open a small parenthesis on a mathematical tool that is **central to them: wavelets**.

We are all familiar with Fourier analysis, which decomposes a signal into sines and cosines. Wavelet analysis is similar, but instead of sines and cosines it uses wavelets: **localised, oscillating functions** that can be **dilated and translated** to analyse the signal at different scales and positions.

For example here you can see the signal being decomposed into wavelet coefficients at different scales. The wavelet coefficient at scale a and position b measures how much structure of size a there is at position b.

[CLICK] This is well suited to convergence maps, which consist of localised structures, each with a characteristic size and position. Convolve the map with a small wavelet and you get its small-scale structure; with a large one, its large-scale structure.

Each band covers a **different range of frequencies**, so we can study the map **scale by scale**, with the bands carrying largely independent information.

`frame 31` · the two statistics · A3.11 · 1:40 · 26:28 in

So the higher-order statistics we will focus on both start from the wavelet decomposision of the map, using an isotropic wavelet **called the starlet**.

And you can see how the starlet transform decomposes the map into **a set of band-pass images**, each carrying structure of a characteristic angular size.

[CLICK] One thing you can do with these band-pass images is to count the peaks on them, the local maxima of the convergence field, and build a histogram of their amplitudes. This is the **wavelet peak counts statistic**.

[CLICK] It is very sensitive to the non-Gaussian features of the field, and it encodes very useful information for cosmology, because as you can imagine, peaks are associated with dark matter haloes, and the abundance and distribution of haloes depends greatly on the cosmological parameters.

[CLICK] But by just counting the peaks, we are throwing away a lot of information: the voids, the filaments, and the rest of the field. So something else you can do is to sum the absolute values of all the wavelet coefficients, in each band, and build a histogram of those sums. This is the **starlet ℓ1-norm statistic**.

[CLICK] You can imagine it as a *PDF of the field in the wavelet domain*.

`frame 32` · from the summary to the parameters · A3.12 · 0:26 · 26:54 in

Ok, so now we have some summary statistics, and we want to compare it with theory predictions we have assuming different cosmological models, to infer the cosmological parameters that best describe the data.

To do that in a way that encodes all the uncertainties that come into play, we use **Bayesian statistics**.

`frame 33` · classical inference · A3.13 · 1:01 · 27:55 in

From Bayes' theorem, the posterior probability of the parameters given the data is **proportional to the likelihood times the prior**: how probable the measured data are for a given cosmology, times what we assumed about the parameters before we looked at the data.

The classical analysis writes that likelihood down **analytically**, usually as a Gaussian in the data vector, which needs a theoretical prediction for the mean and a covariance matrix. For the power spectrum that works, and it is what the field does.

[CLICK] Then you **sample the posterior with MCMC**, and the samples give you the constraints.

For peak counts or the ℓ1-norm we have *neither* an analytic prediction for the mean *nor* a Gaussian likelihood. So we need a different route.

`frame 34` · generative modelling · A3.14 · 0:56 · 28:50 in

The tool **comes from generative modelling**. The problem is this: there is a probability distribution we do not know, and what we have is a set of samples drawn from it.

We want to learn the distribution from the samples, by fitting a parametric model to them.

Once the model is trained we can generate new samples from it, and, also **evaluate its density** -- the probability it assigns to any point.

[CLICK] Generative models are the family of models everyone has heard about in the last few years; they are the algorithms behind image generation, for example. And here you can see how much these models have advanced over the last few years.

`frame 35` · normalizing flows · A3.15 · 0:41 · 29:31 in

The generative model class we use is **the normalizing flow**.

The idea is that you start from a unit Gaussian, and transform it into the target distribution through a series of **simple invertible transformations**, each one parametrised by a neural network.

Because every step is invertible, the model's density follows from the **change-of-variables formula** — and the networks are trained to maximise the likelihood of the training samples.

So *the flow learns from the samples how to turn a Gaussian into the target distribution*.

`frame 36` · simulation-based inference · A3.16 · 0:48 · 30:19 in

So how do we use this for cosmological inference?

While in general we don't have an analytical likelihood, **we have a simulator**: a forward model that we give cosmological parameters and it produces simulated data, with all the stochasticity of the process.

So we draw parameters from the prior, run the simulator on each, and keep the **pairs of parameters and data**.

[CLICK] Those pairs are the training samples for a **conditional normalizing flow**. Trained on them, the flow is *directly a model of the posterior*: we then give it the real observation, and it returns the posterior.

`frame 37` · the divider, Part 3 · A3.16b · 0:17 · 30:37 in

So now that we know how to do all that, we can ask the **third question**: how much of the cosmological information do the different summary statistics keep, and can HOS extract *all* of it?

frame 38 · why build a statistic by hand · A3.17 · 1:11 · 31:48 in

We are looking for the most informative summary statistic for weak lensing. We know the ℓ1-norm extracts more than the power spectrum. [CLICK] But beating the power spectrum is "easy". The real question is **how close to all of the information** in the map a summary can get.

But... since we are already using neural networks for the inference, *why not let one learn the compression as well*?

There are **reasons to be careful**. A network needs a lot of training data. What it learns is hard to interpret. And it generalises less predictably: a feature present in the simulations but not in the real data can bias the result without us noticing.

Nevertheless, people have started doing it, and the theoretically optimal way is to build the network into the simulation-based inference pipeline and train it together with the flow, an idea called **VMIM**.

frame 39 · what optimal means · A3.18 · 0:42 · 32:30 in

The network is a **compressor**: it maps the convergence maps to a low-dimensional summary, and a flow maps the summary to a posterior.

[CLICK] The two are trained together to maximise the **mutual information** between the summary and the parameters, which is equivalent to maximising the expected log posterior.

[CLICK] So a network trained this way in theory estimates *the most information any summary of these maps can carry*.

So what we'd like to do is to compare our analytical HOS to these "optimal" learned summaries.

frame 40 · the setup · A3.19 · 0:37 · 33:07 in

This is the analysis framework we built to do that, and in fact most of the work in this paper is in this figure: building the simulation-based inference pipeline, from the convergence maps to the cosmological parameters, and designing both the analytical statistics and the neural compressor, for which we tested a lot of different architectures, always training with the VMIM objective.

As you see, our two approaches share everything apart from the summary statistic.

frame 41 · the gap · A3.20 · 0:34 · 33:41 in

And here's the first result: the two statistics are not that far apart, but the CNN wins, by **thirty-six per cent** in the figure of merit. So the l1-norm clearly loses a part of the infromation in the data.

[CLICK] However, the comparison is *not symmetric*: there is an important property of these maps that the ℓ1-norm, as we have used it so far, does not take into account.

frame 42 · tomography · A3.21 · 1:01 · 34:42 in

That property hides in **weak-lensing tomography**.

In weak lensing data the source galaxies are sliced into **redshift bins**.

[CLICK] Take the most distant bin.

[CLICK] Its light is lensed by all the matter in front of it,

[CLICK] which gives its convergence map.

[CLICK] A nearer bin

[CLICK] is lensed by a shorter column,

[CLICK] and gives another map.

[CLICK] We get one map per bin, but the maps are *not independent*: the same structure appears in several bins.

How the signal changes from bin to bin is what tells us where the matter sits along the line of sight, and so how structure grew with time.

That is much of the cosmological information in tomography, and a per-bin statistic, which sees each map on its own, cannot see it.

frame 43 · two routes · A3.22 · 1:49 · 36:31 in

With two-point statistics, extending them to capture the cross-bin information is easy: we also compute the two-point functions between different tomographic bins. For higher-order statistics such as the ℓ1-norm there is no obvious equivalent.

We designed two ways to give the ℓ1-norm access to the cross-bin information.

The first option was to create extra maps that capture the "cross" structure: for each pair of tomographic bins, multiply the two maps pixel by pixel. The product is strong only where both maps have structure at the same place. To extract the information from those, we compute the ℓ1-norm on them as well.

[CLICK] The second approach changes the statistic itself. We define a statistic that generalises the ℓ1-norm from a single map to **a pair of tomographic bins**.

Each pixel has a wavelet coefficient in each of the two maps. So instead of one ℓ1 histogram per map, which is what the two curves on the axes are, [CLICK] we build a single **two-dimensional** histogram over the pair, on the grid you see here.

What this encodes is whether a structure that is strong in one map is also strong in the other. **The cells on the diagonal** collect the places where the two maps are equally strong; the cells off the diagonal, the places where one is strong and the other weak.

`frame 44` · the answer · A3.23 · 0:48 · 37:19 in

So let's see what we get with our upgraded higher-order statistics.

Same maps, same flow, four summaries. The per-bin ℓ1-norm. **[CLICK]** Adding the product cross-maps. **[CLICK]** The joint ℓ1-norm. **[CLICK]** And the CNN: which falls almost perfectly onto the joint l1.

So both neural nets and the joint l1 appear to give **the same performance** in terms of extracting cosmological information. And since the network was trained to be information-optimal, this shows that **the joint ℓ1-norm essentially encodes** all the cosmologically useful information in the weak lensing maps.

And it does *without any training*, being computationally efficient, interpretable and robust.

Act 4 — Part 4, baryons · frames 45–50

`frame 45` · the divider · A4.0 · 0:18 · 37:37 in

So now that we have shown how great our higher order statistics are and how well they works in idealized cases, let's go to our final paper to see what happens in a more **messy, realistic scenario**.

`frame 46` · the collision of scales · A4.1 · 2:13 · 39:51 in

Real data have **systematic effects**: things present in the measurement that are not in our model, of instrumental origin, or physical. Not all of them can be modelled, and an effect that is in the data but not in the analysis biases the result. Most groups check a systematic by looking at the data vector. That is not enough: what has to be shown is the effect on the **posterior**.

The dominant astrophysical systematic effect at small scales is **baryonic feedback**: energy from active galactic nuclei and supernovae pushes gas out of haloes and suppresses the matter distribution on small scales. The simulations we can use for inference are dark-matter-only, and the feedback models disagree with each other, so this is physics we cannot model reliably.

The problem is that the non-Gaussian information and the contamination live on **the same small scales**. Since we cannot fully reliably model them for HOS, the conservative approach is to remove those scales, and the question is how impactful this is.

[CLICK] So we may find ourselves either in the *optimistic case*: only the smallest scales are contaminated, and most of the non- Gaussian information survives.

[CLICK] Or in the *pessimistic case*: the contamination reaches much further, and after the cut the power spectrum would do just as well.

So, two questions follow: how exactly does unmodelled feedback bias the higher-order statistics, and how does that grow with area; and once the contaminated scales are cut, is there still a gain over the power spectrum, or all of our efforts in designing better statistics were in vain, cause they are useless in the real world?

`frame 47` · the pipeline · A4.2 · 0:19 · 40:10 in

The pipeline is very similar to the simulation-based one from the previous paper.

[CLICK] simulated maps at different cosmologies, **[CLICK]** Euclid-like noise and masks, **[CLICK]** the starlet transform, **[CLICK]** and the data vector conditions a flow. Same pipeline as before, with one difference.

`frame 48` · how large the bias is · A4.3 · 1:04 · 41:14 in

First question: how large is the bias if we do nothing. CosmoGrid gives every realisation with and without baryonic feedback, so we train the flow on the **dark-matter-only maps**, which is the case of no baryon model at all, and then feed it a **baryonified observation**. We run this for a range of survey areas, to see how the bias evolves as surveys get larger.

For the area of a previous-generation survey the bias is not significant. But the error bars shrink with area while the systematic stays the same, so for a survey like Euclid it becomes *really bad*, and at full sky worse still.

▲ And as expected, the higher-order statistics are more biased than the power spectrum, because they are more sensitive to the contaminated small scales.

`frame 49` · the cuts · A4.4 · 0:38 · 41:52 in

To bring the bias below our threshold of **zero point three sigma**, we remove small-scale information step by step until it is. For the power spectrum, by lowering the maximum multipole, which gives an ell-max for each area, from eight hundred and sixty at two thousand square degrees to three hundred and forty at full sky. For the wavelet statistics, by removing the finest bands, and there dropping **the finest band alone** is enough at every area.

`frame 50` · is there anything left · A4.5 · 1:21 · 43:13 in

So we have seen how the higher-order statistics are biased by unmodelled feedback, and how to cut to remove the bias. The important question is whether anything is left to gain over the power spectrum once the cuts are applied. These are the Stage IV posteriors, on baryon-safe scales only. **[CLICK]** The power spectrum first. **[CLICK]** Peak counts. **[CLICK]** And the ℓ1-norm.

▲ On those scales the ℓ1-norm reaches a figure of merit **one point eight times** the power spectrum's, and at full sky, where the wavelet cut fits better, **two point six**. The peak counts are comparable to the power spectrum, and they carry complementary information, since their degeneracy directions in the w-nought planes differ.

▲ So higher-order statistics are not only deep non-linear probes: the signal survives on *quasi-linear scales*, with no baryon model at all. And in principle we can do better, since our cut is not optimised, and as feedback modelling improves the analysis moves back into the non-linear regime, where the gain is larger.

So, to summarise.

This thesis followed the analysis chain from galaxy shapes to cosmological parameters, and focused mostly on the steps of: making the mass map, extracting information out of it, and doing simulation-based inference.

On the map front, we showed that the choice of mass-mapping method is **not neutral**. It changes the figure of merit significantly. We also developed PnPMass: a method which combines the physics of the problem with deep learning, and that is accurate, with calibrated error bars, and trained once for any mask and noise level.

On the summary statistic, we developed the **joint ℓ1-norm**, and showed it extracts as much cosmological information as a neural compressor trained to be optimal, without any training.

And we showed that higher-order statistics provide significant additional information even after removing the scales contaminated by baryonic feedback.

▲ So, steps of the analysis that are usually treated as neutral choices do change the cosmological result. And with methods that are **calibrated and tested**, we can extract more of the information in the data, and *trust what we get out*.

Thank you.